

Mexi Banco con Docker Compose

Sistemas Distribuidos

Jesus Martínez Trejo

En este laboratorio se realiza el diseño, consumo y validación de la API REST del sistema bancario Mexi-Banco utilizando Postman. Se verifica la correcta atención de las peticiones HTTP, el manejo de errores, las respuestas estructuradas en formato JSON y la automatización de pruebas.

1. Aplicaciones funcionando con docker compose

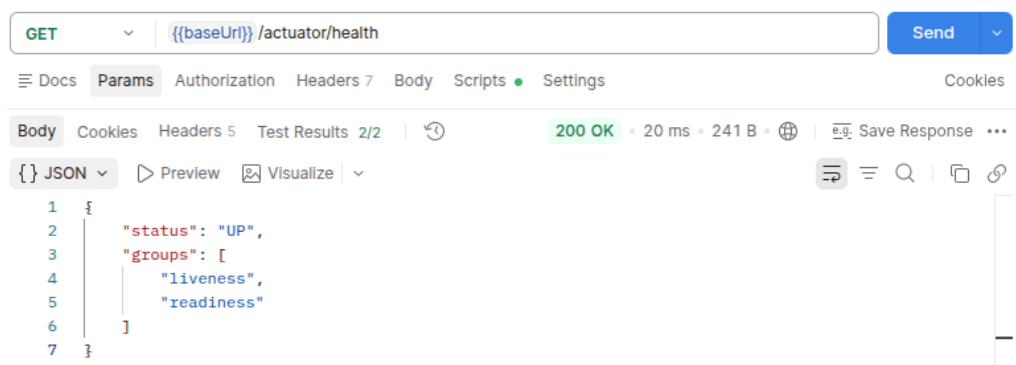
Se creó la aplicación de java y la base de datos con las instrucciones de un docker compose, dónde la app java se construyó a partir del dockerfile que encontró en el mismo ambiente donde se levantaron los servicios.

```
jesus@pc-bdd-jmt:~/mexi-banco$ docker compose ps
NAME                IMAGE              COMMAND                  SERVICE   CREATED         STATUS          PORTS
mexi-banco-app-1    mexi-banco-app    "java -jar app.jar"     app       23 seconds ago  Up 16 seconds  (healthy)   0.0.0.0:8080->8080/tcp, [::]
mexi-banco-db-1     postgres:17-alpine "docker-entrypoint.s..." db        23 seconds ago  Up 22 seconds  (healthy)   5432/tcp
```

2. Evidencia de Pruebas por Endpoint

2.1. Verificación de funcionamiento y prueba inicial

a. Verificación de la salud del sistema



b. Creación de la primera cuenta bancaria

The screenshot shows a REST client interface with a POST request to `{{baseUrl}}/cuentas`. The response is a 201 Created status with a 410 ms response time and 230 B of data. The response body is not visible, but the status bar indicates it was successful.

POST `{{baseUrl}}/cuentas` Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

Body Cookies Headers 5 Test Results 2/2 201 Created • 410 ms • 230 B Save Response

Filter results

PASSED responde 201

PASSED saldo 1000

c. Consultar cuenta creada

The screenshot shows a REST client interface with a GET request to `{{baseUrl}}/cuentas/{{origen}}`. The response is a 200 OK status with a 45 ms response time and 233 B of data. The response body is a JSON object representing a bank account.

GET `{{baseUrl}}/cuentas/{{origen}}` Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Body Cookies Headers 5 Test Results 2/2 200 OK • 45 ms • 233 B Save Response

JSON Preview Visualize

```
1 {
2   "id": 1,
3   "clabe": "002180790563760136",
4   "titular": "Ana",
5   "saldo": 1000.00
6 }
```

2.2. Proceso para una transferencia interna

a. Apertura de una cuenta destino

The screenshot shows a REST client interface with a POST request to `{{baseUrl}}/cuentas`. The response is a 201 Created status with a 30 ms response time and 230 B of data. The response body is a JSON object representing a bank account.

POST `{{baseUrl}}/cuentas` Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

Body 201 Created • 30 ms • 230 B Save Response

JSON Preview Visualize

```
1 {
2   "id": 2,
3   "clabe": "012180790563760136",
4   "titular": "Beto",
5   "saldo": 500
6 }
```

b. Transferencia de \$200

The screenshot shows a REST client interface with a POST request to `{{baseUrl}}/transferencias`. The response is a 201 Created status with a 77 ms latency and 296 B of data. The response body is a JSON object representing a transfer record.

```
POST {{baseUrl}}/transferencias
```

201 Created • 77 ms • 296 B • Save Response

```
{
  "id": 1,
  "claveOrigen": "002180790563760136",
  "claveDestino": "012180790563760136",
  "monto": 200,
  "fecha": "2026-09-28T03:07:03.473790997Z"
}
```

c. Consulta del saldo de la cuenta origen

The screenshot shows a REST client interface with a GET request to `{{baseUrl}}/cuentas/{{origen}}`. The response is a 200 OK status with a 12 ms latency and 232 B of data. The response body is a JSON object representing the account balance.

```
GET {{baseUrl}}/cuentas/{{origen}}
```

200 OK • 12 ms • 232 B • Save Response

```
{
  "id": 1,
  "clabe": "002180790563760136",
  "titular": "Ana",
  "saldo": 800.00
}
```

d. Consulta del saldo de la cuenta destino

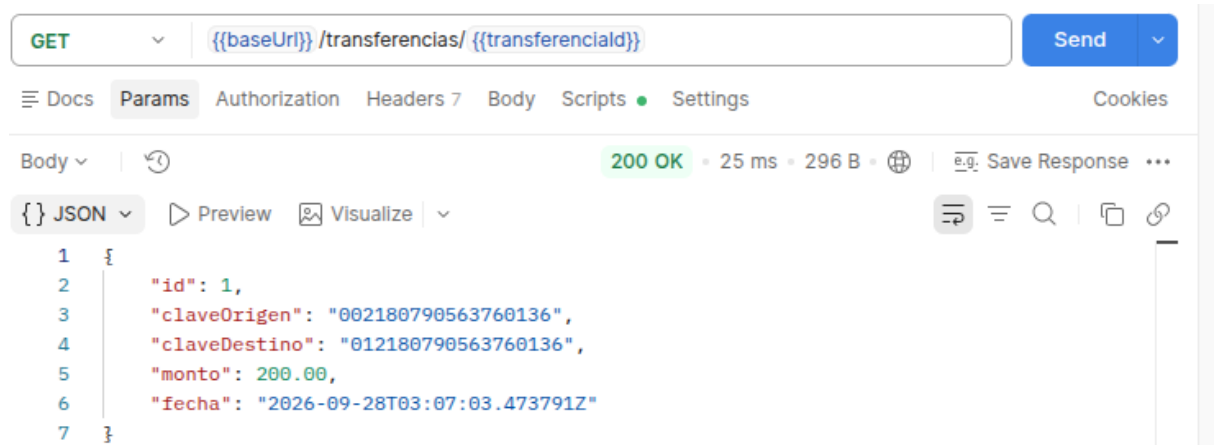
The screenshot shows a REST client interface with a GET request to `{{baseUrl}}/cuentas/{{destino}}`. The response is a 200 OK status with a 14 ms latency and 233 B of data. The response body is a JSON object representing the account balance.

```
GET {{baseUrl}}/cuentas/{{destino}}
```

200 OK • 14 ms • 233 B • Save Response

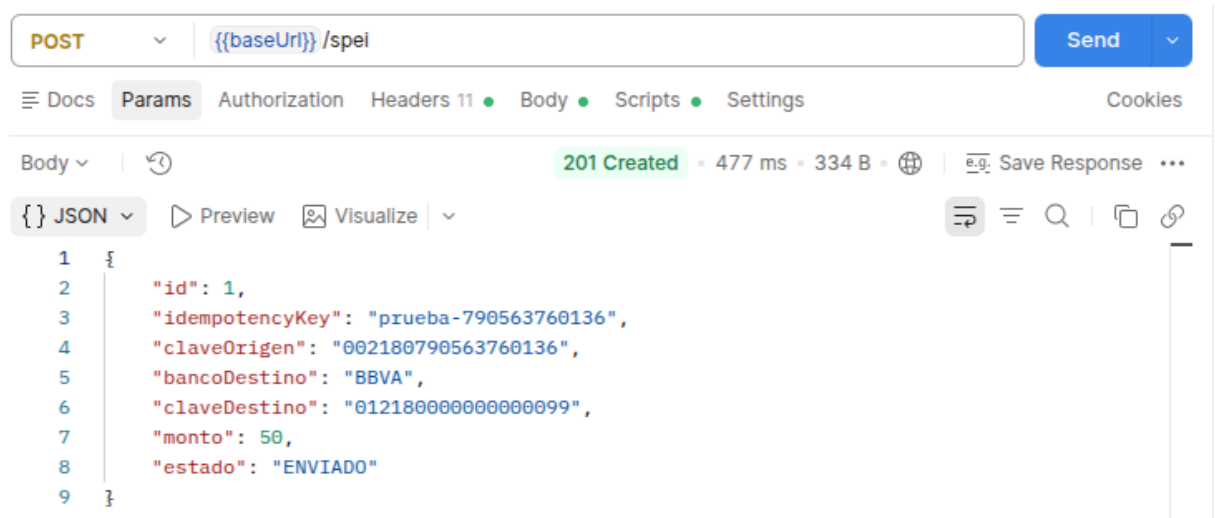
```
{
  "id": 2,
  "clabe": "012180790563760136",
  "titular": "Beto",
  "saldo": 700.00
}
```

e. Consulta de la transferencia

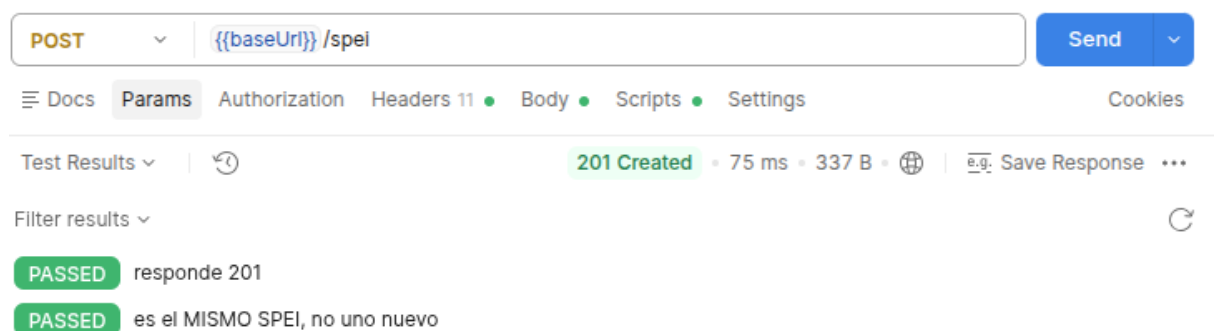


2.3. Transferencia por el sistema SPEI utilizando Idempotency Key

a. Primer intento para transferir \$50 por SPEI



b. Segundo intento para transferir \$50 por SPEI



c. Consulta del saldo de la cuenta origen

GET `{{baseUrl}}/cuentas/{{origen}}` Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Body Cookies Headers 5 Test Results 2/2 200 OK 9 ms 232 B e.g. Save Response ...

{ } JSON Preview Visualize

```
1 {
2   "id": 1,
3   "clabe": "002180790563760136",
4   "titular": "Ana",
5   "saldo": 750.00
6 }
```

2.4. Registro de las operaciones

a. Estados de cuenta origen

GET `{{baseUrl}}/movimientos?clabe={{origen}}` Send

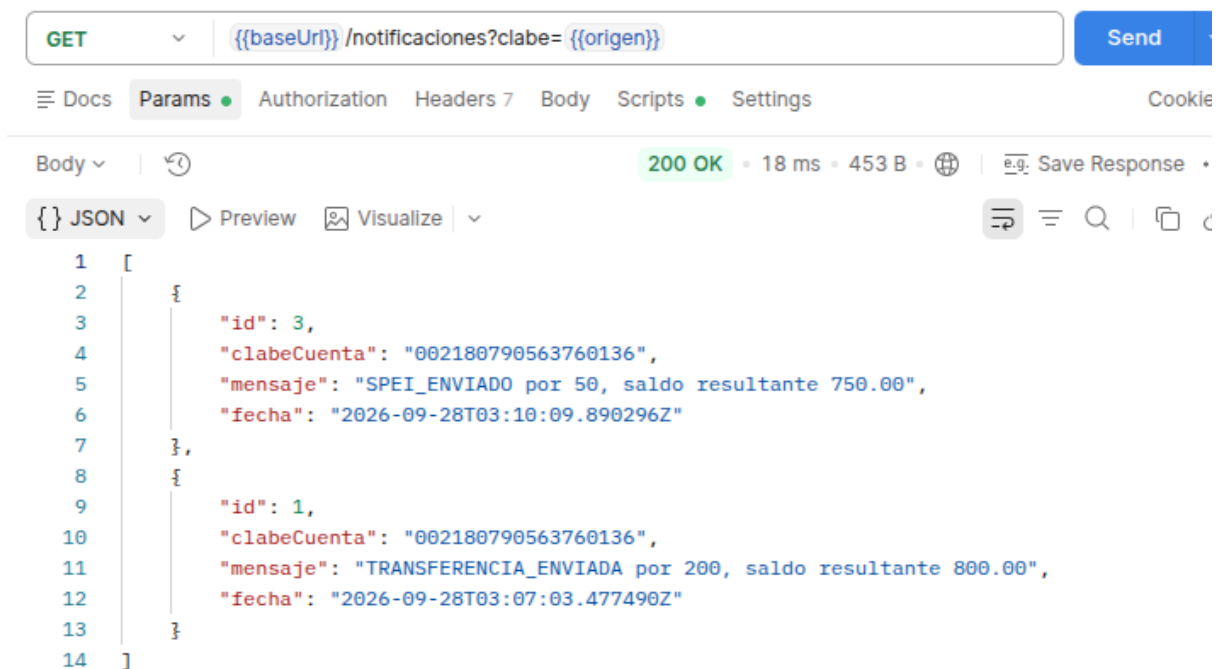
Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Body 200 OK 40 ms 712 B e.g. Save Response ...

{ } JSON Preview Visualize

```
1 [
2   {
3     "id": 5,
4     "clabeCuenta": "002180790563760136",
5     "tipo": "SPEI_ENVIADO",
6     "monto": -50.00,
7     "saldoResultante": 750.00,
8     "fecha": "2026-09-28T03:10:09.477574Z",
9     "referencia": "BBVA - 012180000000000099"
10  },
11  {
12    "id": 3,
13    "clabeCuenta": "002180790563760136",
14    "tipo": "TRANSFERENCIA_ENVIADA",
15    "monto": -200.00,
16    "saldoResultante": 800.00,
17    "fecha": "2026-09-28T03:07:03.470105Z",
18    "referencia": "a 012180790563760136"
19  },
20  {
21    "id": 1,
22    "clabeCuenta": "002180790563760136",
23    "tipo": "DEPOSITO",
24    "monto": 1000.00,
25    "saldoResultante": 1000.00,
26    "fecha": "2026-09-28T02:49:20.817129Z",
27    "referencia": "saldo inicial"
28  }
29 ]
```

b. Avisos de origen



GET `{{baseUrl}}/notificaciones?clabe={{origen}}` Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

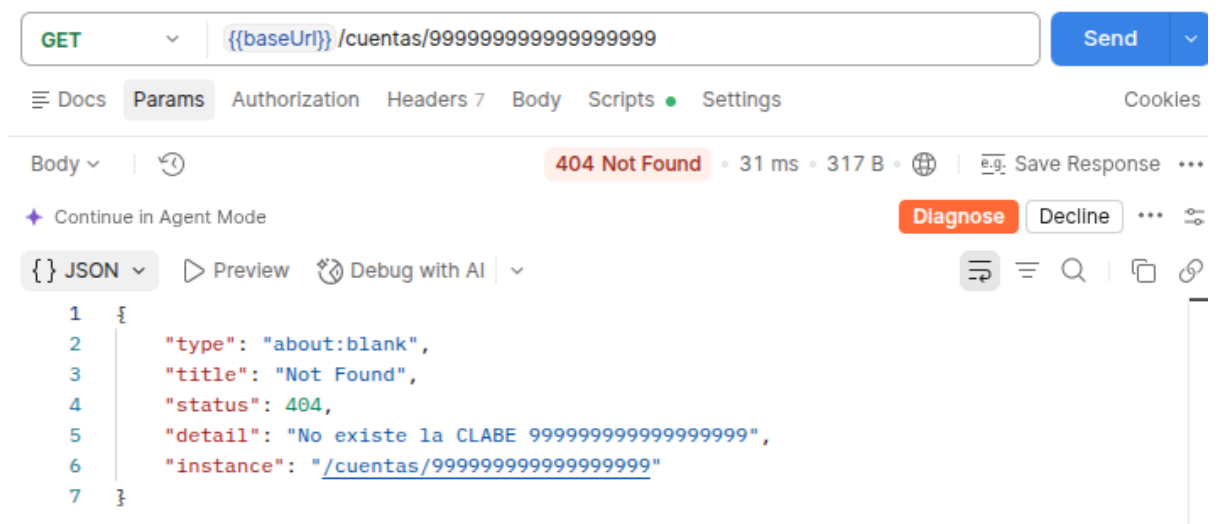
Body 200 OK • 18 ms • 453 B • Save Response

JSON Preview Visualize

```
1  [
2    {
3      "id": 3,
4      "clabeCuenta": "002180790563760136",
5      "mensaje": "SPEI_ENVIADO por 50, saldo resultante 750.00",
6      "fecha": "2026-09-28T03:10:09.890296Z"
7    },
8    {
9      "id": 1,
10     "clabeCuenta": "002180790563760136",
11     "mensaje": "TRANSFERENCIA_ENVIADA por 200, saldo resultante 800.00",
12     "fecha": "2026-09-28T03:07:03.477490Z"
13   }
14 ]
```

2.5. Manejo de errores

a. CLABE que no existe



GET `{{baseUrl}}/cuentas/999999999999999999` Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Body 404 Not Found • 31 ms • 317 B • Save Response

Continue in Agent Mode Diagnose Decline

JSON Preview Debug with AI

```
1  {
2    "type": "about:blank",
3    "title": "Not Found",
4    "status": 404,
5    "detail": "No existe la CLABE 999999999999999999",
6    "instance": "/cuentas/999999999999999999"
7  }
```

b. Intento de crear cuenta con CLABE repetida

The screenshot shows a REST client interface with a POST request to `{{baseUrl}}/cuentas`. The response is a 409 Conflict with a status of 409, a title of "Conflict", and a detail message: "Ya existe una cuenta con la CLABE 002180790563760136". The instance is `/cuentas`. The response body is JSON:

```
1 {
2   "type": "about:blank",
3   "title": "Conflict",
4   "status": 409,
5   "detail": "Ya existe una cuenta con la CLABE 002180790563760136",
6   "instance": "/cuentas"
7 }
```

c. Intento de crear cuenta sin CLABE

The screenshot shows a REST client interface with a POST request to `{{baseUrl}}/cuentas`. The response is a 400 Bad Request with a status of 400, a title of "Bad Request", and a detail message: "Add missing request body". The instance is `/cuentas`. The response body is JSON:

```
1 {
2   "timestamp": "2026-09-28T03:14:14.976+00:00",
3   "status": 400,
4   "error": "Bad Request",
5   "path": "/cuentas"
6 }
```

d. Intento de transferir más dinero del que se tiene

The screenshot shows a REST client interface with a POST request to `{{baseUrl}}/transferencias`. The response is a 422 Unprocessable Entity (WebDAV) (RFC 4918) with a status of 422, a title of "Unprocessable Entity", and a detail message: "Saldo insuficiente en la CLABE 002180790563760136". The instance is `/transferencias`. The response body is JSON:

```
1 {
2   "type": "about:blank",
3   "title": "Unprocessable Entity",
4   "status": 422,
5   "detail": "Saldo insuficiente en la CLABE 002180790563760136",
6   "instance": "/transferencias"
7 }
```

e. SPEI sin Idempotency-Key

The screenshot shows a REST client interface with a POST request to `{{baseUrl}}/spei`. The response is a 400 Bad Request with a status of 400, a message of "Bad Request", and a path of `/spei`. The response body is a JSON object:

```
1 {
2   "timestamp": "2026-09-28T03:15:00.837+00:00",
3   "status": 400,
4   "error": "Bad Request",
5   "path": "/spei"
6 }
```

f. Estado de la cuenta origen creada previamente

The screenshot shows a REST client interface with a GET request to `{{baseUrl}}/cuentas/{{origen}}`. The response is a 200 OK with a status of 200, a message of "OK", and a body of 232 B. The response body is a JSON object:

```
1 {
2   "id": 1,
3   "clabe": "0602180790563760136",
4   "titular": "Ana",
5   "saldo": 750.00
6 }
```

g. Intento de transferirse a sí mismo

The screenshot shows a REST client interface with a POST request to `{{baseUrl}}/transferencias`. The response is a 422 Unprocessable Entity (WebDAV) (RFC 4918) with a status of 422, a message of "Unprocessable Entity", and a detail of "La cuenta origen y la destino son la misma". The response body is a JSON object:

```
1 {
2   "type": "about:blank",
3   "title": "Unprocessable Entity",
4   "status": 422,
5   "detail": "La cuenta origen y la destino son la misma",
6   "instance": "/transferencias"
7 }
```

3. Prueba de persistencia de datos

Para comprobar el funcionamiento de los servicios y el aislamiento de datos, se realizaron pruebas apagando el entorno con y sin destrucción del volumen `mexibanco_datos`.

- Con docker compose down los datos son persistentes.

```
jesus@pc-bdd-jmt:~/mexi-banco$ docker compose down
[+] down 3/3
✓ Container mexi-banco-app-1 Removed
✓ Container mexi-banco-db-1 Removed
✓ Network mexibanco Removed
jesus@pc-bdd-jmt:~/mexi-banco$ docker compose up -d
[+] up 3/3
✓ Network mexibanco Created
✓ Container mexi-banco-db-1 Healthy
✓ Container mexi-banco-app-1 Started
```

The screenshot shows a REST client interface with a GET request to `{{baseUrl}}/cuentas/{{origen}}`. The response is a 200 OK status with a response time of 216 ms and a body size of 232 B. The response body is in JSON format, showing account details for ID 1, including a clabe, titular name, and balance.

```
{
  "id": 1,
  "clabe": "002180790563760136",
  "titular": "Ana",
  "saldo": 750.00
}
```

- Con docker compose down -v se eliminan los registros de la base de datos.

```
jesus@pc-bdd-jmt:~/mexi-banco$ docker compose down -v
[+] down 4/4
✓ Container mexi-banco-app-1      Removed
✓ Container mexi-banco-db-1      Removed
✓ Volume mexi-banco_mexibanco_datos Removed
✓ Network mexibanco              Removed
jesus@pc-bdd-jmt:~/mexi-banco$ docker compose up -d
[+] up 4/4
✓ Network mexibanco              Created
✓ Volume mexi-banco_mexibanco_datos Created
✓ Container mexi-banco-db-1      Healthy
✓ Container mexi-banco-app-1     Started
```

GET Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Body 404 Not Found • 188 ms • 317 B • Save Response

JSON Preview Debug with AI

```
1 {
2   "type": "about:blank",
3   "title": "Not Found",
4   "status": 404,
5   "detail": "No existe la CLABE 002180790563760136",
6   "instance": "/cuentas/002180790563760136"
7 }
```